Statistical Physics

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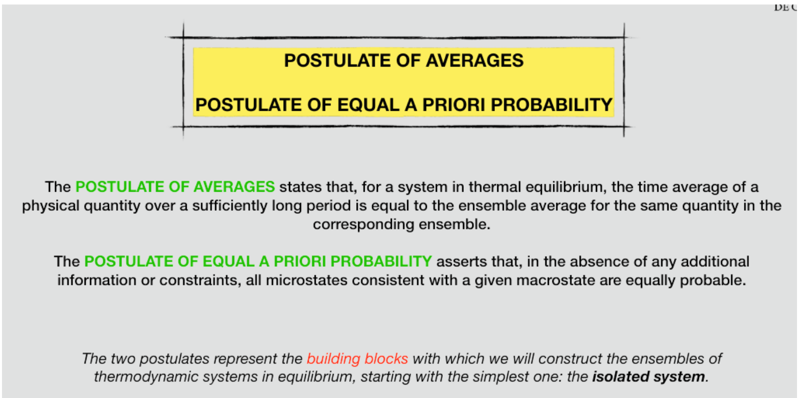
Índice general

Capítulo 1

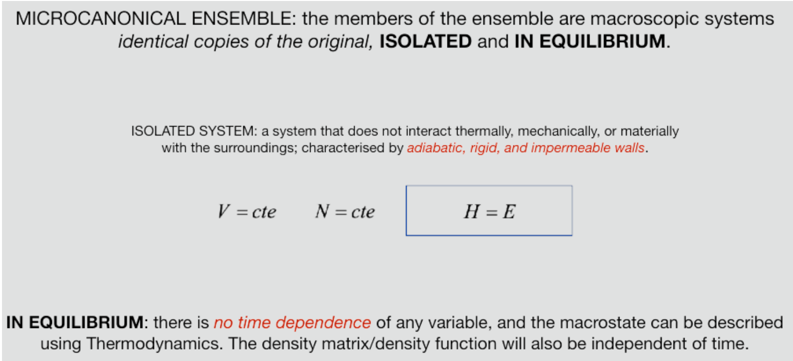
Microcanonical Ensemble

1.1. Introduction

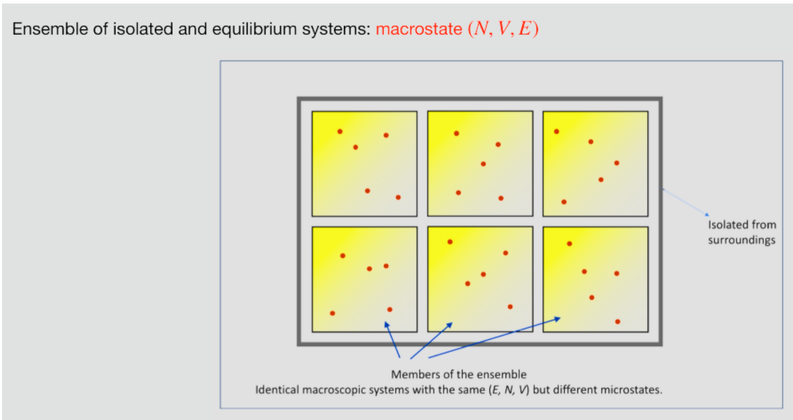


We have stressed in the *Introduction* that a macroscopic system consists of approximately $N = 10^{23}$ particles and, correspondingly, exhibits an energy spectrum that varies within $\Delta E \sim e^{-N}$. Any attempt to find a solution of the equations of motion at the microscopic level is hopeless and also superfluous as during the measurement process (which takes a finite amount of time) both macroscopic and microscopic properties of the system change. We have also learnt that a many-body system cannot be characterised by a single microstate, but rather by an **ensemble of microstates** which is determined by macroscopic variables $(E, V, N, \dots)$. Experience shows that, at sufficiently long time scales, any microscopic system evolves towards an equilibrium state, in which

$$\frac{\partial \hat{\rho}}{\partial t} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}] = 0 \tag{1.1}$$

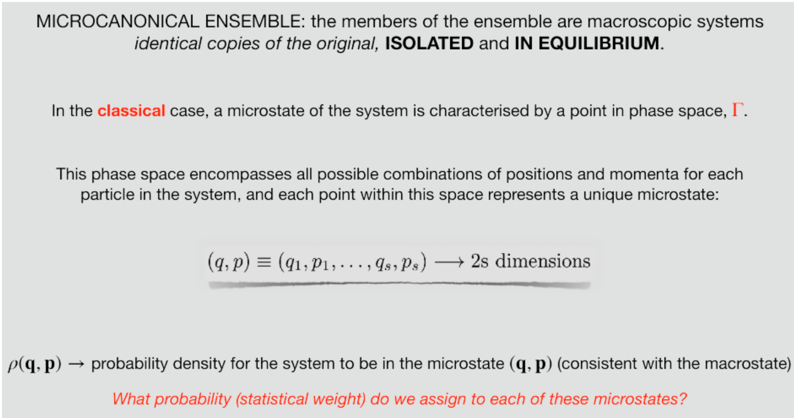


must be satisfied. Since, according to Eq. (1.1), the density operator commuted with the Hamiltonian operator, in an equilibrium ensemble ρ can depend only on conserved quantities, the microscopic properties of the system change continually even at equilibrium, but the distribution of microstates within the ensemble does not depend on time. In this chapter, to fix these ideas, we will introduce the microcanonical ensemble, where number of particles (N), system volume (V) and energy (E) are constant. To this end, we will make use of the *Postulate of Averages* and *Postulate of Equal A Priori Probability*. These postulates are the building blocks with which we will construct the thermodynamic ensembles in equilibrium, starting with the simplest: the isolated system.



The members of the microcanonical ensemble are macroscopic systems identical to the original, **isolated**, and **in equilibrium**. An isolated system does not interact thermally, mechanically, or materially with the outside; it has adiabatic, rigid, and impermeable walls. As such, it exhibits constant volume, constant number of particles. Additionally $H = E$. Systems in equilibrium have no temporal dependencies on any variable, and their macrostate can be described using thermodynamics. The density matrix/function also does not depend on time.

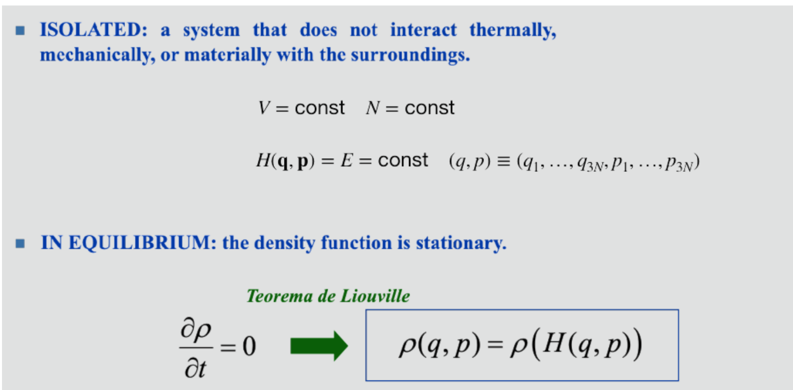
1.2. Classical Microcanonical Ensemble



In the microcanonical ensemble, the constituents of the ensemble represent macroscopic systems that are exact copies of the original system. These systems are kept in isolation, do not interact with the external environment, and are maintained in a state of thermodynamic equilibrium. In the classical context, a microstate of the system is distinctly characterised by a specific point within the system's phase space, denoted as Γ. This phase space **encompasses all possible combinations of positions and momenta for each particle in the system, and each point within this space represents a unique microstate**:

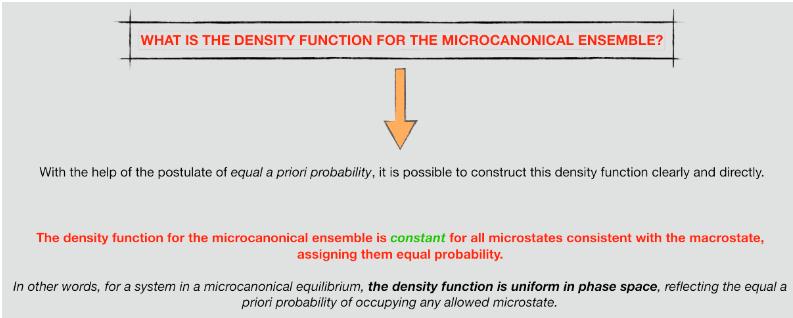
$$(\mathbf{q}, \mathbf{p}) \equiv (q_1, p_1, \ldots, q_s, p_s) \longrightarrow 2s \text{ dimensions} \tag{1.2}$$

The probability density associated with the likelihood of the system being in a given microstate, $\rho(q, p)$, characterises the relative probability of finding the system in that specific microstate defined by its generalised coordinates q and momenta p within the phase space. In a sense, it quantifies the "crowding" of microstates within the phase space, indicating which microstates are more or less probable within the constraints of the given macrostate.



What probability (statistical weight) do we assign to each of these microstates?

We know that our system is isolated and in equilibrium. This implies that N, V and E are constant and the density function does not change with time. Now, what is the density function for the microcanonical ensemble?



With the help of the postulate of equal a priori probability, it is possible to construct this density function clearly and directly. In the microcanonical ensemble, all microstates compatible with a particular macrostate have the same probability of occurring, simplifying the construction of the density function. Generally speaking:

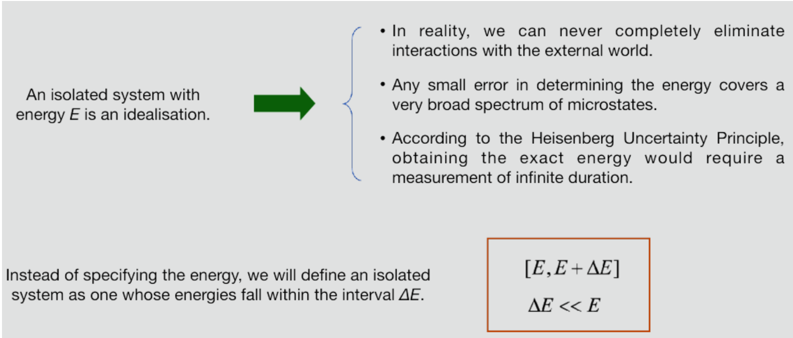
$$\rho(\mathbf{q}, \mathbf{p}) = \begin{cases} \text{constant} & \text{if } (\mathbf{q}, \mathbf{p}) \text{ is compatible with the macrostate} \\ 0 & \text{otherwise} \end{cases} \tag{1.3}$$

The constant in this function is chosen in such a way that the **sum or integral of the density function over the entire phase space** (q, p) **equals 1**, ensuring that the total probability of finding the system in some microstate within the macrostate is 1. Let's now add some specific details:

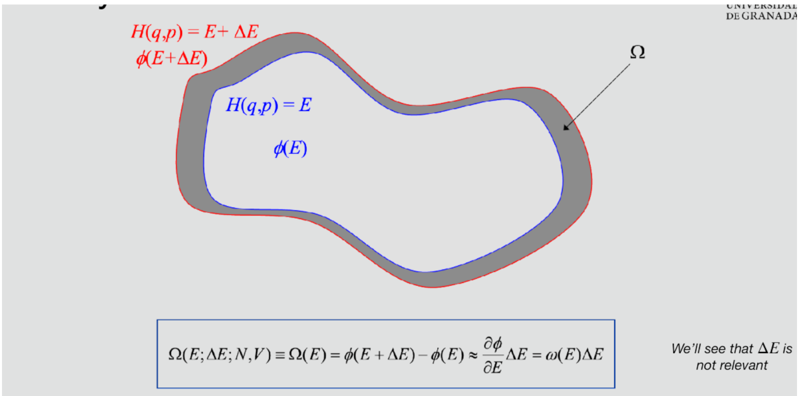
1. In any average within this ensemble, it must be taken into account that all mechanical states (q, p) for which $H \neq E$ cannot contribute to the mean value, as they do not conserve energy (see Fig. 4.4).
2. ρ is exclusively a function of H.
3. Equal a priori probability for all microstates with $H = E$.
4. Normalization condition:

$$\int \rho(\mathbf{q}, \mathbf{p}) \, d\mathbf{q} \, d\mathbf{p} = 1. \tag{1.4}$$

1.2.1. Density of States



An isolated system with a precisely defined energy E is an idealisation. In reality, we can never completely eliminate interactions with the external world. Any small error in determining the energy covers an extremely wide spectrum of microstates. According to the *Heisenberg Uncertainty Principle*, obtaining the energy exactly would require a measurement of infinite duration. As such, instead of specifying the energy precisely, we define an isolated system as one whose energies fall within the interval $[E, E + \Delta E]$ with $\Delta E \ll E$.



In particular, we consider a small but finite shell $[E, E+\Delta E]$ close to the energy surface. With reference to Fig. 4.3, $\Omega(N, V, E)$ is the volume of the shell bounded by the two energy surfaces with energies E and $E + \Delta E$ or, equivalently, the volume of phase space accessible to a system with N particles, volume V, and total energy E. In other words, $\Omega(N, V, E)$ quantifies the number of microstates available to the system under those conditions:

$$\Omega(N, V, E) = \int_{E \leq H \leq E+\Delta E} d\mathbf{q}\, d\mathbf{p} \tag{1.5}$$

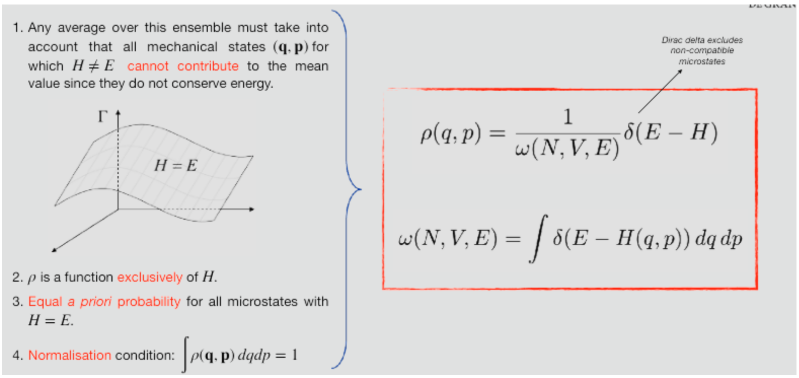
Let now ϕ be the total volume of phase space enclosed by the energy surface E. We then have that

$$\Omega(N, V, E) = \phi(E + \Delta E) - \phi(E) \tag{1.6}$$

Taking the limit for $\Delta E \to 0$, corresponding to an infinitesimal shell, one obtains

$$\Omega(N, V, E) \approx \frac{\partial \phi(E)}{\partial E} \Delta E \equiv \omega \Delta E, \quad \Delta E \ll E \tag{1.7}$$

$\Omega(N, V, E)$ refers to the *microcanonical ensemble multiplicity*. It represents the number of microstates that correspond to a specific energy E in a microcanonical ensemble. In other words, $\Omega(N, V, E)$ quantifies the **number of distinct ways in which the energy E can be distributed among the particles in the system** while satisfying the fixed constraints of N and V.



From Ω and its dependence on the variables N, V, and E, the entire thermodynamics of the microcanonical ensemble can be deduced. Additionally, the normalisation constant

$$\omega(N, V, E) = \frac{\partial \phi(E)}{\partial E} = \int \delta(E - H(\mathbf{q}, \mathbf{p})) d\mathbf{q}\, d\mathbf{p} \tag{1.8}$$

represents the **density of states**, indicating the number of compatible microstates per unit energy interval. By contrast, Ω gives the total volume of phase space accessible to the system, quantifying the number of microstates available under specified conditions.

Let's see how the density of states is related to the phase-space density $\rho(\mathbf{q}, \mathbf{p})$, which describes how microstates are distributed in phase space (remember this is a probability!). In the microcanonical ensemble, all accessible states are equally probable, meaning:

$$\rho(\mathbf{q}, \mathbf{p}) \propto \delta(H(\mathbf{q}, \mathbf{p}) - E) \tag{1.9}$$

Since the system is equally likely to be in any of these microstates, the normalisation factor must ensure that:

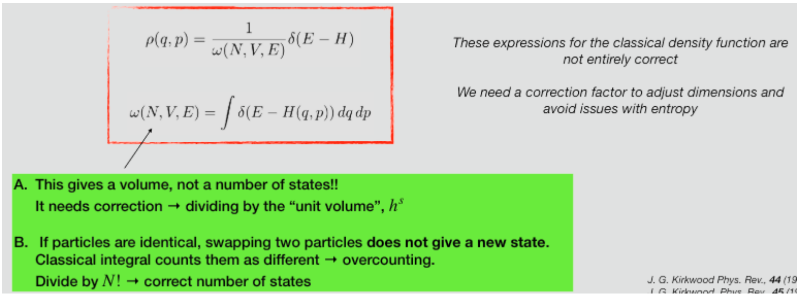
$$\int \rho(\mathbf{q}, \mathbf{p})\, d\mathbf{q}\, d\mathbf{p} = 1 \tag{1.10}$$

Using the definition of $\omega(E)$, we then set:

$$\rho(\mathbf{q}, \mathbf{p}) = \frac{\delta(H(\mathbf{q}, \mathbf{p}) - E)}{\omega(E)} \tag{1.11}$$

where the Dirac delta excludes non-compatible microstates. Equivalently,

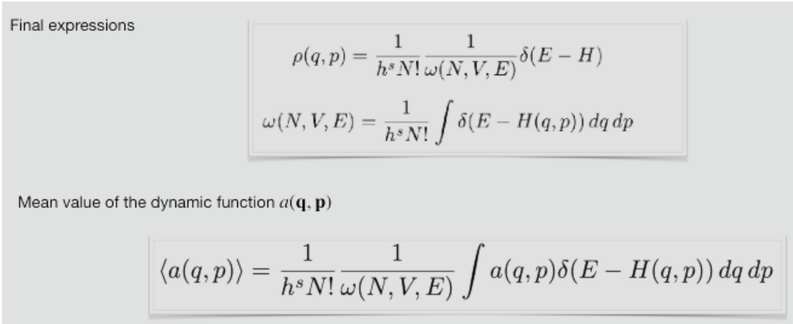
$$\rho(\mathbf{q}, \mathbf{p}) = \begin{cases} \frac{1}{\omega(E)} & E \leq H(\mathbf{q}, \mathbf{p}) \leq E + \Delta E \\ 0 & \text{otherwise} \end{cases} \tag{1.12}$$



The expressions given above for the classical density function provide a useful framework for understanding the statistical behaviour of macroscopic systems in the classical realm. However, it's important to note that these classical expressions may not hold true when we transition to the quantum realm. The behaviour of particles at the quantum level is inherently probabilistic and governed by the Heisenberg uncertainty principle, which is encapsulated in h. Boltzmann realised that the particles in the system are actually indistinguishable. This implies dividing the number of states $\omega(E)$ by an additional $N!$, thereby accounting for the number of different ways to label the particles. Additionally, to satisfy dimensional requirements, Boltzmann suggested to use a constant that has units of momentum times length, which we now know to be h^s, with h the Planck constant and s the number of degrees of freedom of the system. The interested reader is referred to McQuarrie, p. 114 for further details [3]. In particular,

$$\Delta q_i \Delta p_i \sim h \tag{1.13}$$

$$\Delta q^{(s)} \Delta p^{(s)} \sim h^{(s)} \tag{1.14}$$



We therefore end up with the following final expressions:

$$\rho(\mathbf{q}, \mathbf{p}) = \frac{1}{h^s N!} \frac{1}{\omega(N,V,E)} \delta(E - H) \tag{1.15}$$

$$\omega(N,V,E) = \frac{1}{h^s N!} \int \delta(E - H(\mathbf{q}, \mathbf{p}))\, d\mathbf{q}\, d\mathbf{p} \tag{1.16}$$

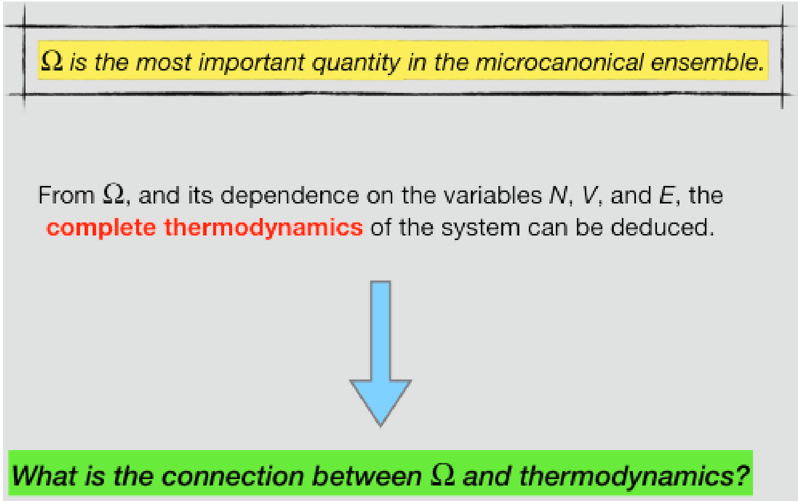
The mean value of the dynamic function is given by:

$$\langle a(\mathbf{q}, \mathbf{p}) \rangle = \frac{1}{h^s N!} \frac{1}{\omega(N,V,E)} \int a(\mathbf{q}, \mathbf{p}) \delta(E - H(\mathbf{q}, \mathbf{p}))\, d\mathbf{q}\, d\mathbf{p} \tag{1.17}$$

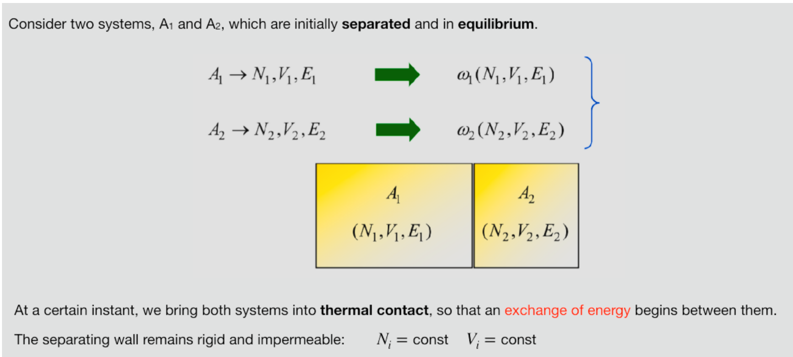
and the number of states with energy in the interval $[0, E]$ reads

$$\phi(N,V,E) \equiv \phi(E) = \frac{1}{h^s N!} \int \theta(E - H)\, d\mathbf{q}\, d\mathbf{p}, \quad \text{with } \theta(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases} \tag{1.18}$$

where θ is the Heaviside step function.



1.3. Connection with Thermodynamics



Let's consider two systems, A_1 and A_2, which are initially separated and in equilibrium, as shown in Fig. above. At a certain moment, we bring both systems into thermal contact, allowing an exchange of energy between them. The separating wall remains rigid and impermeable: $N_i = \text{const}, V_i = \text{const}$. However, the walls allows for heat exchange: $E_i \neq \text{const}$. The global system $A = A_1 + A_2$ is also isolated. Its total energy reads $E = E_1 + E_2 + E_{12}$, with E_{12} being negligible compared to E_1 and E_2, $E \approx E_1 + E_2$.

$$\left.\begin{array}{lcl} A_1 \to N_1, V_1, E_1 & \Longrightarrow & \omega_1(N_1, V_1, E_1) \\ A_2 \to N_2, V_2, E_2 & \Longrightarrow & \omega_2(N_2, V_2, E_2) \end{array}\right\}$$

The coordinates and moment of A_1 and A_2 are, respectively, $(\mathbf{q}_1, \mathbf{p}_1)$ and $(\mathbf{q}_2, \mathbf{p}_2)$, whereas those of the global system A are $(\mathbf{q}, \mathbf{p})$. It follows that the global Hamiltonian can be expressed as

$$H(\mathbf{q}, \mathbf{p}) = H_1(\mathbf{q}_1, \mathbf{p}_1) + H_2(\mathbf{q}_2, \mathbf{p}_2) \tag{1.19}$$

Now, let's calculate the number of accessible states of the isolated system $A = A_1 + A_2$. The number of microstates that correspond to energy E, which is the multiplicity introduced in Eq. 4.5, reads

$$\Omega(N,V,E) = \omega(N,V,E)\Delta E, \tag{1.20}$$

with

$$\omega(N,V,E) = \frac{1}{h^{s_1+s_2} N_1! N_2!} \int \delta(E - H(q,p))\, dq\, dp \tag{1.21}$$

This equation can be re-written by incorporating the linearity of the Hamiltonian:

$$\omega(E) = \frac{1}{h^{s_1} N_1!} \int dq_1\, dp_1 \frac{1}{h^{s_2} N_2!} \int dq_2\, dp_2\, \delta(E - \underbrace{H_1(q_1,p_1)}_{E_1} - H_2(q_2,p_2)) \underbrace{\int_0^E \delta(E_1 - H_1(q_1,p_1))\, dE_1}_{=1} \tag{1.22}$$

the last integral in the right hand side of the above equation has just been added for reasons that become clear in the following equation

$$\omega(E) = \int_0^E dE_1 \underbrace{\frac{1}{h^{s_1} N_1!} \int dq_1\, dp_1\, \delta(E_1 - H_1(q_1,p_1))}_{\omega_1(E_1)} \underbrace{\frac{1}{h^{s_2} N_2!} \int dq_2\, dp_2\, \delta(E - E_1 - H_2(q_2,p_2))}_{\omega_2(E-E_1)} \tag{1.23}$$

It follows that

$$\omega(E) = \int_0^E \omega_1(E_1)\omega_2(E - E_1)\, dE_1 \tag{1.24}$$

The energy E of the global system is fixed, but E_1 can vary. Only the values of E_1 that are sufficiently close to E_1^* contribute to the integral. If $|E_1 - E_1^*| \gg 0$, then the number of microstates decreases abruptly. What is the value of E_1 that is compatible with equilibrium? If we refer to it as E_1^*, it follows that the corresponding equilibrium value for A_2 will be $E_2^* = E - E_1^*$. The above integral $\omega(E)$ (see red curve in Fig. 4.6) tends to a distribution that is especially peaked at E_1^*. The maximum of this distribution indicates the equilibrium macrostate with largest probability of being observed. Let's approximate $\omega(E)$ to a Gaussian distribution:

$$\omega_1(E_1)\omega_2(E - E_1) \simeq \omega_1(E_1^*)\omega_2(E - E_1^*) e^{-(E_1 - E_1^*)^2/2\sigma_E^2} \tag{1.25}$$

The idea behind this approximation is to assume that around the point E_1^*, the densities $\omega_1(E_1)$ and $\omega_2(E-E_1)$ can be approximated by their values at E_1^*, with a Gaussian çorrection"term to account for variations away from E_1^*. This is often used in statistical mechanics when combining two densities of states that are expected to behave similarly near their maximum values (see Fig. 4.6).

It follows that

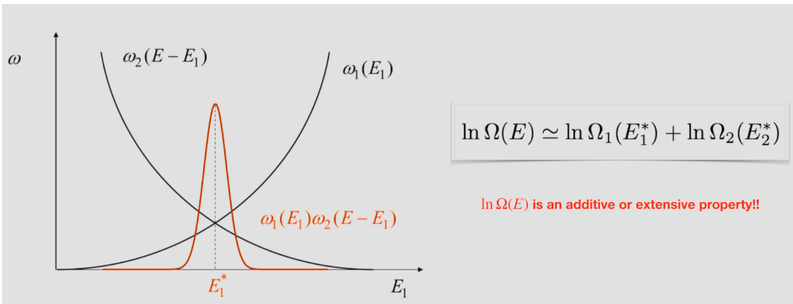
$$\omega(E) = \int_0^E \omega_1(E_1)\omega_2(E-E_1)\,dE_1 \simeq \omega_1(E_1^*)\omega_2(E-E_1^*)\int_0^E e^{-(E_1-E_1^*)^2/2\sigma_E^2}\,dE_1 \tag{1.26}$$

The integral between 0 and E can be approximated to the integral between $-\infty$ and $+\infty$, that is $\int_0^E \simeq \int_{-\infty}^{+\infty}$. We then note that

$$\int_{-\infty}^{+\infty} e^{-(E_1-E_1^*)^2/2\sigma_E^2}\,dE_1 = \sqrt{2\pi}\sigma_E \tag{1.27}$$

Therefore

$$\omega(E) \simeq \omega_1(E_1^*)\omega_2(\underbrace{E-E_1^*}_{E_2^*})\sqrt{2\pi}\sigma_E \tag{1.28}$$



If we now multiply both sides by ΔE and take the logarithm, we obtain

$$\omega(E)\Delta E = \Omega(E) \simeq \omega_1(E_1^*)\Delta E\omega_2(E_2^*)(\Delta E)^2\frac{\sqrt{2\pi}\sigma_E}{\Delta E} = \Omega_1(E_1^*)\Omega_2(E_2^*)\frac{\sqrt{2\pi}\sigma_E}{\Delta E} \tag{1.29}$$

$$\ln\Omega(E) \simeq \ln\Omega_1(E_1^*) + \ln\Omega_2(E_2^*) + \ln\left(\frac{\sqrt{2\pi}\sigma_E}{\Delta E}\right) \tag{1.30}$$

Some observations:

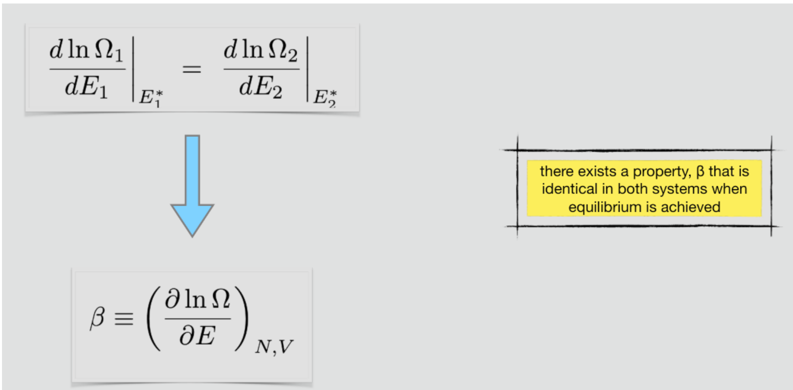
- $\Omega(E)$ increases exponentially with N in *normal* systems; $\ln(\Omega(E))$ increases linearly with N; $\sqrt{\sigma_E}$ grows as $\sqrt{N}$; and ΔE does not change.
- $\ln(\sigma_E/\Delta E)$ grows as $\ln(N)$ and in the limit of macroscopic systems $\ln(N) \ll N$ and $N \gg 1$.

We therefore conclude that

$$\ln \Omega(E) \simeq \ln \Omega_1(E_1^*) + \ln \Omega_2(E_2^*) \tag{1.31}$$

This result is especially important as it implies that $\ln \Omega(E)$ is an additive or **extensive** property.

1.3.1. Condition of Thermal Equilibrium



What are the conditions observed when equilibrium is achieved? In other words, what do we observe when $A_1 \to E_1^*$ and $A_2 \to E_2^*$? Let's locate the maximum of the distribution by differentiating the distribution with respect to E_1:

$$\frac{d}{dE_1}[\omega_1(E_1)\omega_2(E - E_1)]_{E_1^*} = 0 \tag{1.32}$$

It follows that

$$\left.\frac{d\omega_1(E_1)}{dE_1}\right|_{E_1^*} \omega_2(E - E_1^*) + \omega_1(E_1^*) \left.\frac{d\omega_2(E - E_1)}{dE_1}\right|_{E_1^*} = 0 \tag{1.33}$$

Since $E_2 = E - E_1$, then $d/dE_2 = -d/dE_1$ and the above equation can be rewritten as

$$\left.\frac{d\omega_1(E_1)}{dE_1}\right|_{E_1^*} \omega_2(E_2^*) - \omega_1(E_1^*) \left.\frac{d\omega_2(E_2)}{dE_2}\right|_{E_2^*} = 0 \tag{1.34}$$

Equivalently:

$$\left.\frac{1}{\omega_1}\frac{d\omega_1}{dE_1}\right|_{E_1^*} = \left.\frac{1}{\omega_2}\frac{d\omega_2}{dE_2}\right|_{E_2^*} \tag{1.35}$$

which gives

$$\left.\frac{1}{\Omega_1}\frac{d\Omega_1}{dE_1}\right|_{E_1^*} = \left.\frac{1}{\Omega_2}\frac{d\Omega_2}{dE_2}\right|_{E_2^*} \tag{1.36}$$

and equivalently

$$\underbrace{\left.\frac{d\ln\Omega_1}{dE_1}\right|_{E_1^*}}_{\text{depends on } A_1} = \underbrace{\left.\frac{d\ln\Omega_2}{dE_2}\right|_{E_2^*}}_{\text{depends on } A_2} \tag{1.37}$$

We have therefore proved that there exists a property, β that is identical in both systems when equilibrium is achieved:

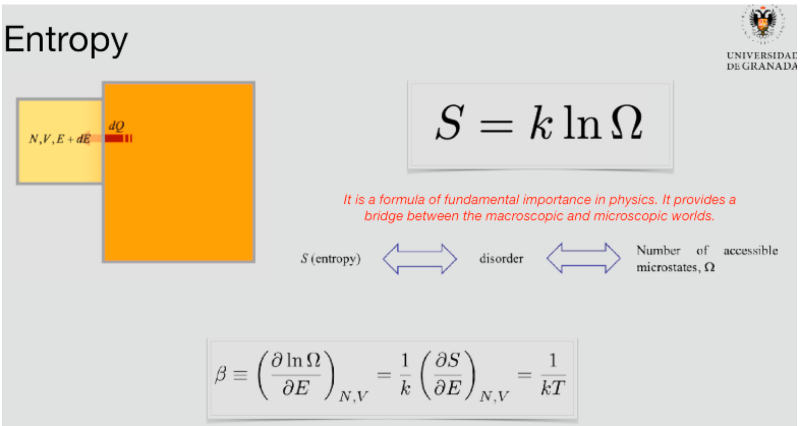
$$\beta \equiv \left(\frac{\partial \ln\Omega}{\partial E}\right)_{N,V} \tag{1.38}$$

We can finally obtain the *condition of thermal equilibrium*:

$$\beta_1 = \beta_2 \tag{1.39}$$

This property is necessarily a function of intensive variables only, that is $\beta = \beta(T, \rho, \mu, \dots)$, as it does not depend on the system size. Nevertheless, we still have to understand what $\ln\Omega$ and β actually are and to what thermodynamic properties they are associated.

1.3.2. Entropy



Let's write its total derivative as follows

$$dX = \left(\frac{\partial \ln \Omega}{\partial E}\right)_{N,V} dE \tag{4.40}$$

Therefore

$$dE = \frac{dX}{\left(\frac{\partial \ln \Omega}{\partial E}\right)_{N,V}} \tag{4.41}$$

From the First and Second Law of Thermodynamics, we know that

$$dQ = TdS = dE + pdV \tag{1.40}$$

Because volume is constant

$$dE = \frac{dX}{\left(\frac{\partial \ln \Omega}{\partial E}\right)_{N,V}} = TdS \tag{1.41}$$

We can then write dS as the total differential of X - it only depends on X

$$dS = \frac{dX}{T\left(\frac{\partial \ln \Omega}{\partial E}\right)_{N,V}} \tag{1.42}$$

Such an exclusive dependence implies that the factor accompanying dX can only depend on X. Mathematically:

$$dS = \frac{dX}{f(X)} \tag{1.43}$$

By integrating the above equation, we obtain

$$S = \underbrace{\int \frac{dX}{f(X)}}_{g(X)} + S_0 \tag{1.44}$$

We know that entropy is an extensive property:

$$S_{1+2} = S_1 + S_2 \tag{1.45}$$

and so is $X = \ln \Omega$:

$$X_{1+2} = X_1 + X_2 \tag{1.46}$$

Therefore

$$S_{1+2} = g(X_{1+2}) + S_{0,1+2} = g(X_{1+2}) + S_{0,1} + S_{0,2} = g(X_1 + X_2) + S_{0,1} + S_{0,2} \tag{1.47}$$

It is also true that

$$S_{1+2} = S_1 + S_2 = g(X_1) + S_{0,1} + g(X_2) + S_{0,2} \tag{1.48}$$

It follows that

$$g(X_1 + X_2) = g(X_1) + g(X_2) \tag{1.49}$$

The only function that satisfies the above equation is a linear function of X:

$$g(X) = kX \tag{1.50}$$

where k is a constant. Therefore

$$S = kX + S_0 \tag{1.51}$$

By substituting the X function with its original value, we get:

$$S = k \ln \Omega + S_0 \tag{1.52}$$

In the limit of a temperature approaching zero ($T \to 0$ K), there exist only one accessible microstate, which is the one with the lowest possible energy. It then follows that, for this particular situation, $\Omega = 1$:

$$\lim_{T \to 0} S = k \ln 1 + S_0 = S_0 \tag{1.53}$$

The Third Law of Thermodynamics states that the entropy of a closed system at equilibrium approaches a constant value when its temperature approaches absolute zero. This constant value, here denoted as S_0, is usually set to zero by convention. It then follows that

$$S = k \ln \Omega \tag{1.54}$$

This is a formula of fundamental importance in physics. It provides a bridge between the macroscopic and microscopic worlds. Now we know that $\ln \Omega$ refers to. What about β?

$$\beta \equiv \left(\frac{\partial \ln \Omega}{\partial E} \right)_{N,V} = \frac{1}{k} \left(\frac{\partial S}{\partial E} \right)_{N,V} = \frac{1}{kT} \tag{1.55}$$

It is then clear that β is the *inverse temperature*, whereas k is a *universal constant* that we don't know yet. It is also clear why Eq. 4.39 gives $\beta_1 = \beta_2$ for two systems in contact that achieve equilibrium: their temperature must be the same. Now, how do we calculate the constant k? Since k is a constant, its value is expected to be the same regardless the system. To calculate it, we can then make use of the simplest thermodynamic system, that is the classical monotonic ideal gas. The equation of state of this system is known:

$$pV = nRT \tag{1.56}$$

where $n = N/N_A$ is the number of moles, T the absolute temperature, p pressure, V volume and $R \simeq 8{,}314$ J Kmol^{-1} the ideal-gas constant. In the next section, we will use this gas to derive an expression of entropy that is functional to understand what k is and calculate its numerical value.

1.4. Classical Monoatomic Ideal Gas

$$H = \sum_{j=1}^{N} \left[\frac{p_j^2}{2m} + U_r(\bar{r}_j) \right], \; U_r(\bar{r}_j) = \begin{cases} 0 & \text{if } \bar{r}_j \leq r \\ \infty & \text{if } \bar{r}_j > r \end{cases}$$

$$\boxed{S(N,V,E) \simeq kN \left\{ \ln \left[\frac{V}{N} \left(\frac{4\pi m E}{3h^2 N} \right)^{3/2} \right] + \frac{5}{2} \right\}}$$

Sackur-Tetrode formula (1912)

Let's write the Hamiltonian of a monoatomic ideal gas enclosed in a volume V of radius r:

$$H = \sum_{j=1}^{N} \left[\frac{p_j^2}{2m} + U_r(\bar{r}_j) \right] \tag{4.59}$$

To calculate the entropy S of this gas, we need to know the number of microstates:

$$\Omega(N, V, E; \Delta E) \longrightarrow S = k \ln \Omega \tag{4.60}$$

Let's introduce $\phi(E)$, which is the number of microstates with energy in the interval $[0, E]$. Calculating $\phi(E)$ is easier than calculating the total number of compatible microstates with a macrostate of energy E, that is $\Omega(E)$. In particular:

$$\phi(N,V,E) = \frac{1}{h^{3N} N!} \int \theta(E-H) dq\, dp = \frac{1}{h^{3N} N!} \int \theta(E-H) d\bar{r}_1 \ldots d\bar{r}_N d\bar{p}_1 \ldots d\bar{p}_N \tag{4.61}$$

Therefore

$$\phi(N,V,E) = \frac{1}{h^{3N} N!} \int d\bar{r}_1 \ldots d\bar{r}_N \int d\bar{p}_1 \ldots d\bar{p}_N \theta \left[E - \sum_{j=1}^{N} \left(\frac{p_j^2}{2m} + U_r(\bar{r}_j) \right) \right] \tag{4.62}$$

If we pay attention to the above integral, we see that the function $\theta(E - H)$ is zero for all points behind radius r and, for all the remaining points $U_r(\bar{r}_j) = 0$. Consequently, Eq. 4.62 can be simplified and re-written as follows

$$\phi(N,V,E) = \frac{1}{h^{3N}N!} \underbrace{\int d\bar{r}_1 \ldots d\bar{r}_N}_{\text{accessible volume}} \underbrace{\int d\bar{p}_1 \ldots d\bar{p}_N \, \theta \left[E - \sum_{j=1}^{N} \left(\frac{p_j^2}{2m} \right) \right]}_{\text{independent of } \{\bar{r}_j\}} \tag{4.63}$$

We then obtain

$$\phi(N,V,E) = \frac{V^N}{h^{3N}N!} \int dp_1 \ldots dp_{3N} \, \theta \left[E - \sum_{j=1}^{3N} \left(\frac{p_j^2}{2m} \right) \right] \tag{4.64}$$

Because $\theta(x-a) = \theta(b(x-a))$, with b positive constant, then

$$\phi(N,V,E) = \frac{V^N}{h^{3N}N!} \underbrace{\int dp_1 \ldots dp_{3N} \, \theta \left[2mE - \sum_{j=1}^{3N} p_j^2 \right]}_{\text{volume of a 3N-D hypersphere of } r=\sqrt{2mE}} \tag{4.65}$$

Only momenta that satisfy $\sum_{j=1}^{3N} p_j^2 \leq 2mE$ contribute to the above integral. For the volume of a hypersphere see the note below. In particular

Nota

The general formula for the d-dimensional volume of a hypersphere reads

$$V_d(r) = r^d \frac{\pi^{d/2}}{\Gamma\left(\frac{d}{2}+1\right)} \tag{4.66}$$

For example, if $d = 2$, we recover the area of a circle:

$$V_2(r) = r^2 \frac{\pi}{\Gamma(2)} = \pi r^2. \tag{1.57}$$

If $d = 3$, then

$$V_3(r) = r^3 \frac{\pi^{3/2}}{\Gamma(5/2)} = r^3 \frac{\pi^{3/2}}{\frac{3}{2}\frac{1}{2}\Gamma(1/2)} = \frac{4}{3} r^3 \frac{\pi^{3/2}}{\pi^{1/2}} = \frac{4}{3}\pi r^3. \tag{1.58}$$

We also remind that

$$\Gamma(n+1) = n! \tag{1.59}$$

$$\phi(N,V,E) = \frac{V^N}{h^{3N}N!}(2mE)^{3N/2}\frac{\pi^{3N/2}}{\Gamma\left(\frac{3N}{2}+1\right)} = \frac{V^N}{h^{3N}N!}\frac{(2\pi mE)^{3N/2}}{\Gamma\left(\frac{3N}{2}+1\right)} \tag{1.60}$$

We can finally estimate the number of states in the interval $[E, E+\Delta E]$:

$$\Omega(E;\Delta E) = \frac{\partial\phi}{\partial E}\Delta E = \frac{V^N}{h^{3N}N!}\frac{(2\pi mE)^{3N/2}}{\Gamma\left(\frac{3N}{2}+1\right)}\frac{3N}{2}\frac{\Delta E}{E} \tag{1.61}$$

This equation allows us to calculate entropy, $S = k\ln\Omega$, in the thermodynamic limit, that is for $N \to \infty$, $V \to \infty$ and $E \to \infty$:

$$S(N,V,E) = \tag{1.62}$$

$$= k\ln\Omega = k\left\{N\ln\left[\frac{V(2\pi mE)^{3/2}}{h^3}\right] - \ln N! - \ln\Gamma\left(\frac{3N}{2}+1\right) + \ln\left(\frac{3N}{2}\right) + \ln\frac{\Delta E}{E}\right\} = \tag{1.63}$$

$$= k\left\{N\ln\left[\frac{V(2\pi mE)^{3/2}}{h^3}\right] - \ln N! - \left(\frac{3N}{2}+1\right)\ln\left(\frac{3N}{2}+1\right) + \left(\frac{3N}{2}+1\right) + \ln\left(\frac{3N}{2}\right) \tag{1.64}$$

$$\simeq k\left\{N\ln\left[\frac{V(2\pi mE)^{3/2}}{h^3}\right] - \ln N! - \left(\frac{3N}{2}\right)\ln\left(\frac{3N}{2}\right) + \left(\frac{3N}{2}\right) + \ln\left(\frac{3N}{2}\right) + \ln\frac{\Delta E}{E}\right\} \tag{1.65}$$

where we used that $\ln\Gamma(x) \simeq x\ln x - x$. We now apply Stirling's approximation[4] and get

$$S(N,V,E) = kN\ln\left[\frac{V(2\pi mE)^{3/2}}{h^3}\right] + \tag{1.66}$$

$$+ k\left\{-N\ln N + N - \frac{3N}{2}\ln\left(\frac{3N}{2}\right) + \frac{3N}{2} + \ln\left(\frac{3N}{2}\right) + \ln\frac{\Delta E}{E}\right\} = \tag{1.67}$$

$$= kN\ln\left[\frac{V(2\pi mE)^{3/2}}{h^3}\right] + k\left\{-N\ln N + N - \left(\frac{3N}{2}-1\right)\ln\left(\frac{3N}{2}\right) + \frac{3N}{2} + \ln\frac{\Delta E}{E}\right\} \simeq \tag{1.68}$$

$$\simeq kN\ln\left[\frac{V(2\pi mE)^{3/2}}{h^3}\right] + kN\left\{-\ln N + 1 - \ln\left(\frac{3N}{2}\right)^{3/2} + \frac{3}{2} + \frac{1}{N}\ln\frac{\Delta E}{E}\right\} \simeq \tag{1.69}$$

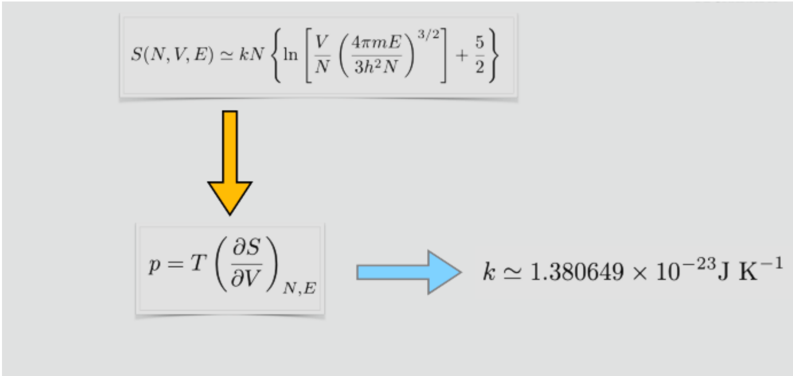
$$\simeq kN\left\{\ln\left[\frac{V}{N}\frac{(2\pi mE)^{3/2}}{h^3}\right] - \ln\left(\frac{3N}{2}\right)^{3/2} + \frac{5}{2}\right\} \tag{1.70}$$

and merging the two logarithms, we finally get

$$S(N,V,E) \simeq kN\left\{\ln\left[\frac{V}{N}\left(\frac{4\pi mE}{3h^2N}\right)^{3/2}\right] + \frac{5}{2}\right\} \tag{4.74}$$

Eq. 4.74, referred to as the "Sackur-Tetrode formula", gives the absolute entropy of a classical monoatomic ideal gas. Note that both V/N and E/N in the above equation are **intensive properties**. As such, the absolute entropy can be concisely written as $S/kN = f(\text{intensive})$, where $f(\text{intensive})$ indicates that S/kN is a function of intensive properties. Nevertheless, $S \propto Nf(\text{intensive})$ is an **extensive property** as it increases with the number of particles in the system.

Boltzmann Constant



We are now in the conditions to obtain the numerical value of the constant k. To this end, we need to calculate the pressure of the ideal gas

$$p = T\left(\frac{\partial S}{\partial V}\right)_{N,E} = TkN\frac{\partial}{\partial V}\ln V = TkN\frac{1}{V} \tag{1.71}$$

Rearranging, we obtain

$$pV = NkT \tag{4.76}$$

and because the ideal-gas law $pV = nRT$ must hold, we conclude that

$$Nk = nR \tag{4.77}$$

equivalently

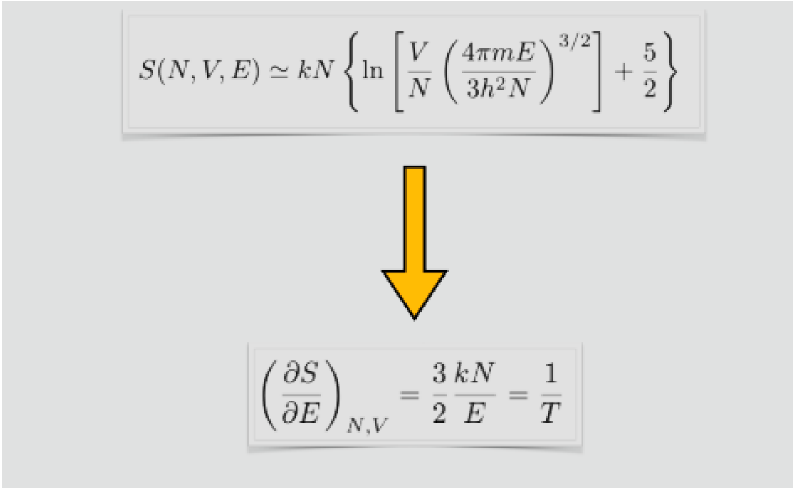
$$k = \frac{n}{N}R = \frac{R}{N_A} \tag{4.78}$$

where N_A is the Avogadro's number and R the ideal-gas constant. Therefore:

$$k \simeq 1{,}380649 \times 10^{-23}\text{J K}^{-1} \tag{4.79}$$

This constant is known as the **Boltzmann constant**, named after the Austrian physicist Ludwig Boltzmann, who made significant contributions to the development of statistical mechanics and the understanding of the behaviour of particles in a gas.

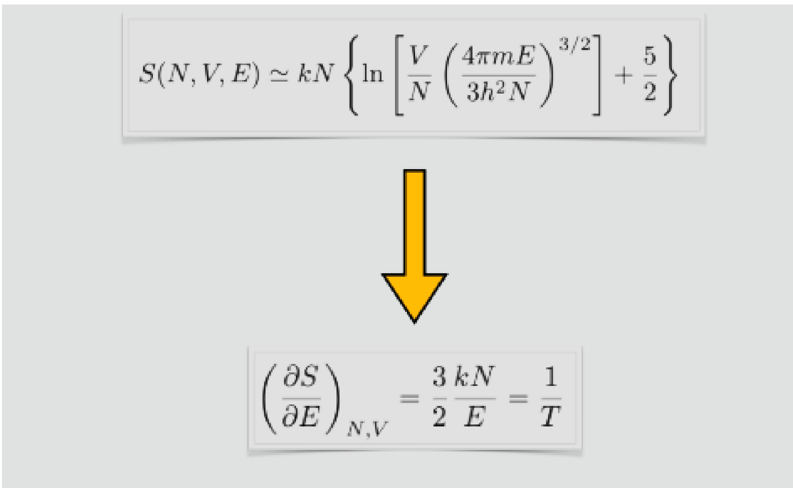
Intensive and Extensive Properties from Entropy



Let's differentiate the Sackur-Tetrode equation with respect to E, V and N, the three key properties of the microcanonical ensemble:

$$\left(\frac{\partial S}{\partial E} \right)_{N,V} = \frac{3}{2} \frac{kN}{E} = \frac{1}{T} \tag{1.72}$$

where we have applied the equipartition theorem.


$S(N,V,E) \simeq kN \left\{ \ln \left[\frac{V}{N} \left(\frac{4\pi mE}{3h^2N} \right)^{3/2} \right] + \frac{5}{2} \right\}$
$\left(\frac{\partial S}{\partial E} \right)_{N,V} = \frac{3}{2} \frac{kN}{E} = \frac{1}{T}$

Additionally, differentiating S with respect to V, one obtains

$$\left(\frac{\partial S}{\partial V} \right)_{N,E} = \frac{kN}{V} = \frac{p}{T} \tag{1.73}$$

where we have applied the ideal-gas law $pV = NkT$. Finally, we can define the *chemical potential* as

$$\mu = -T\left(\frac{\partial S}{\partial N}\right)_{V,E} = -kT\ln\left[\frac{V}{N}\left(\frac{4\pi mE}{3h^2N}\right)^{3/2}\right] = -kT\ln\left[\frac{V}{N}\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\right] \tag{1.74}$$

where we have used $E/N = 3/2kT$. We can now write down the total derivative of S with respect to N, V and E:

$$dS = \frac{1}{T}dE + \frac{p}{T}dV - \frac{\mu}{T}dN \tag{1.75}$$

or

$$dE = TdS - pdV + \mu dN \tag{1.76}$$

It is then possible to calculate the intensive variables in terms of energy change:

$$T = \left(\frac{\partial E}{\partial S}\right)_{N,V} \tag{4.85}$$

$$p = -\left(\frac{\partial E}{\partial V}\right)_{N,S} = T\left(\frac{\partial S}{\partial V}\right)_{N,E} \tag{4.86}$$

$$\mu = \left(\frac{\partial E}{\partial N}\right)_{V,S} = -T\left(\frac{\partial S}{\partial N}\right)_{E,V} \tag{1.77}$$

From here, we can also obtain the heat capacities at constant volume and constant pressure

$$C_v = \left(\frac{\partial E}{\partial T}\right)_{N,V} = T\left(\frac{\partial S}{\partial T}\right)_{N,V} = \frac{3}{2}kN \tag{1.78}$$

where we have again made use of the equipartition theorem.

$$C_p = \left(\frac{\partial H}{\partial T}\right)_{N,P} = T\left(\frac{\partial S}{\partial T}\right)_{N,P} = \frac{5}{2}kN \tag{1.79}$$

where H stands for enthalpy and we have made use of the ideal-gas law.

We stress that both properties are extensive. Nevertheless, the heat capacities per mol, that is $c_v = C_v/n$ and $c_p = C_p/n$, usually referred to as specific heats, are intensive properties. We can then obtain the Mayer relation:

$$c_p - c_v = \frac{5}{2}\frac{kN}{n} - \frac{3}{2}\frac{kN}{n} = R \tag{1.80}$$

Following similar steps, we can obtain the Gibbs free energy (G), enthalpy (H) and Helmholtz free energy (A):

$$G = A + pV = \mu N = -NkT \ln\left[\frac{V}{N}\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\right] \tag{1.81}$$

$$H = E + pV = G + TS = \frac{5}{2}NkT \tag{1.82}$$

$$A = E - TS = \mu N - pV = -NkT\left\{\ln\left[\frac{V}{N}\left(\frac{2\pi mkT}{h^2}\right)^{3/2}\right] + 1\right\} \tag{1.83}$$